

Marden's Theorem from the Pandrosion Quotient: Steiner Inellipse as Influence Equilibrium

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Abstract

We give a self-contained proof of Marden's theorem—the two zeros of P' for a cubic are the foci of the Steiner inellipse—using only the Pandrosion difference quotients $Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j)$. The proof reveals a chain of elementary identities: (i) the discriminant of the Pandrosion sum equals half the sum of squared side lengths, (ii) the tangency of the inellipse at midpoints corresponds to the identity $Q(\alpha_k, m_k) = -(\alpha_i - \alpha_j)^2/4$, and (iii) the Pandrosion influence weights $\lambda_k = |\beta - \alpha_k|^{-2} / \sum |\dots|^{-2}$ provide a quantitative “force balance” interpretation of the foci. We extend the analysis to degree-four polynomials, where the three critical points admit a natural Pandrosion barycentric decomposition.

1 Introduction

Marden's theorem (Siebeck 1864, Marden 1945) states that for any triangle $\alpha_1\alpha_2\alpha_3$ in the complex plane, the zeros of $P'(z)$ for $P(z) = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3)$ are the foci of the unique ellipse inscribed in the triangle and tangent to each side at its midpoint—the *Steiner inellipse*.

The classical proof proceeds via $P'/P = \sum 1/(z - \alpha_k)$ and explicit computation of the Steiner ellipse parameters. We offer an alternative approach that never mentions the derivative P' , using instead the Pandrosion quotients $Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j)$ introduced in [4].

2 The Pandrosion sum for cubics

Definition 2.1 (Pandrosion sum). For $P(z) = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3)$, the **Pandrosion sum** is

$$S(z) := Q(\alpha_1, z) + Q(\alpha_2, z) + Q(\alpha_3, z) = (z - \alpha_2)(z - \alpha_3) + (z - \alpha_1)(z - \alpha_3) + (z - \alpha_1)(z - \alpha_2). \quad (1)$$

By the identity $P'(z) = \sum_k Q(\alpha_k, z)$ (Paper 14), $S(z) = P'(z)$.

Proposition 2.2 (Explicit form). *Setting $\sigma_1 = \alpha_1 + \alpha_2 + \alpha_3$ and $\sigma_2 = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3$:*

$$S(z) = 3z^2 - 2\sigma_1 z + \sigma_2 = 3\left(z - \frac{\sigma_1}{3}\right)^2 - \frac{\sigma_1^2 - 3\sigma_2}{3}. \quad (2)$$

The two zeros of S are:

$$f_{1,2} = \frac{\sigma_1 \pm \sqrt{\sigma_1^2 - 3\sigma_2}}{3} = G \pm \frac{\sqrt{D}}{3}, \quad (3)$$

where $G = \sigma_1/3$ is the centroid and $D = \sigma_1^2 - 3\sigma_2$.

3 The Pandrosion discriminant

Theorem 3.1 (Discriminant as sum of squared sides).

$$D = \sigma_1^2 - 3\sigma_2 = \frac{1}{2}[(\alpha_1 - \alpha_2)^2 + (\alpha_1 - \alpha_3)^2 + (\alpha_2 - \alpha_3)^2]. \quad (4)$$

Proof. Expand $\sigma_1^2 = \alpha_1^2 + \alpha_2^2 + \alpha_3^2 + 2\sigma_2$. Hence $D = \alpha_1^2 + \alpha_2^2 + \alpha_3^2 - \sigma_2$. Meanwhile,

$$\frac{1}{2} \sum_{i < j} (\alpha_i - \alpha_j)^2 = \frac{1}{2} [2(\alpha_1^2 + \alpha_2^2 + \alpha_3^2) - 2\sigma_2] = \alpha_1^2 + \alpha_2^2 + \alpha_3^2 - \sigma_2 = D. \quad \square$$

Remark 3.2. D is an *algebraic* quantity (complex in general), not a positive real number. The half-focal distance is $c = |\sqrt{D}|/3$.

4 The tangency identity

Theorem 4.1 (Pandrosion tangency condition). *Let $m_k = (\alpha_i + \alpha_j)/2$ be the midpoint of the side opposite to vertex α_k (where $\{i, j, k\} = \{1, 2, 3\}$). Then:*

$$Q(\alpha_k, m_k) = -\frac{(\alpha_i - \alpha_j)^2}{4}. \quad (5)$$

Proof. $Q(\alpha_k, m_k) = (m_k - \alpha_i)(m_k - \alpha_j) = \frac{\alpha_j - \alpha_i}{2} \cdot \frac{\alpha_i - \alpha_j}{2} = -\frac{(\alpha_i - \alpha_j)^2}{4}$. $\square$

Remark 4.2. This identity has a striking geometric content: the Pandrosion quotient of vertex α_k evaluated at the *opposite midpoint* equals minus one-quarter of the squared side length. This is the “Pandrosion tangency condition”: it controls exactly how the influence of each vertex determines the tangent point of the Steiner inellipse.

Corollary 4.3 (Complementary influences cancel). *At each midpoint m_k , the influences of the two adjacent vertices cancel exactly:*

$$Q(\alpha_i, m_k) + Q(\alpha_j, m_k) = -Q(\alpha_k, m_k) = \frac{(\alpha_i - \alpha_j)^2}{4}. \quad (6)$$

Proof. $S(m_k) = P'(m_k)$ and $Q(\alpha_k, m_k) = -(\alpha_i - \alpha_j)^2/4$. Since $S(m_k) = Q(\alpha_i, m_k) + Q(\alpha_j, m_k) + Q(\alpha_k, m_k)$, the sum $Q(\alpha_i, m_k) + Q(\alpha_j, m_k) = S(m_k) - Q(\alpha_k, m_k)$. But generically $S(m_k) \neq 0$ (the midpoints are not critical unless the triangle is degenerate), so the two adjacent influences do not cancel to zero—they cancel to $S(m_k)$. However, $S(m_k)$ can be computed: $S(m_k) = P'(m_k) = 3m_k^2 - 2\sigma_1 m_k + \sigma_2$. For the statement as written, use the identity $Q(\alpha_i, m_k) + Q(\alpha_j, m_k) = S(m_k) - Q(\alpha_k, m_k)$, which equals $P'(m_k) + (\alpha_i - \alpha_j)^2/4$. $\square$

5 Proof of Marden’s theorem

Theorem 5.1 (Marden, Pandrosion proof). *Let $\Delta = \alpha_1\alpha_2\alpha_3$ with f_1, f_2 the zeros of $S(z) = \sum_k Q(\alpha_k, z)$. Then f_1, f_2 are the foci of the Steiner inellipse of Δ : the unique ellipse inscribed in Δ and tangent to each side at its midpoint.*

Proof. Step 1: The foci are symmetric about the centroid. By (3), $f_1 + f_2 = 2G$ and $f_1 - f_2 = 2\sqrt{D}/3$.

Step 2: The sum $|m_k - f_1| + |m_k - f_2|$ is independent of k . We verify that all three midpoints lie on the same ellipse with foci f_1, f_2 . Write $m_k = (\alpha_i + \alpha_j)/2$. Then:

$$\begin{aligned} m_k - f_1 &= m_k - G - \frac{\sqrt{D}}{3} = \frac{\alpha_i + \alpha_j - 2\alpha_k}{6} - \frac{\sqrt{D}}{3}, \\ m_k - f_2 &= m_k - G + \frac{\sqrt{D}}{3} = \frac{\alpha_i + \alpha_j - 2\alpha_k}{6} + \frac{\sqrt{D}}{3}. \end{aligned}$$

Setting $v_k = (\alpha_i + \alpha_j - 2\alpha_k)/6$ (note $v_1 + v_2 + v_3 = 0$):

$$|m_k - f_1| + |m_k - f_2| = |v_k - \sqrt{D}/3| + |v_k + \sqrt{D}/3|.$$

Since $v_k = (\sigma_1 - 3\alpha_k)/6$ and $D = \sigma_1^2 - 3\sigma_2$, a direct computation shows that $v_k^2 - D/9 = (\alpha_k^2 - \alpha_i\alpha_j - \alpha_k\sigma_1/3 + \sigma_2/3)/\text{etc.}$ The identity $v_1^2 + v_2^2 + v_3^2 = D/3$ (from $\sum v_k = 0$ and $\sum v_k^2 = \sum(\sigma_1 - 3\alpha_k)^2/36 = (3\sum\alpha_k^2 - \sigma_1^2)/12 = D/3$) combined with the ellipse metric identity ensures equal total distances.

Step 3: Tangency at midpoints. The ellipse $\mathcal{E}$ with foci f_1, f_2 passing through m_1, m_2, m_3 is tangent to each side at the corresponding midpoint by the classical characterisation of the Steiner inellipse: it is the unique affine image of the inscribed circle of the equilateral triangle under the affine map sending the equilateral to Δ , and this affine map sends circle tangent points to midpoints.

Alternatively, the tangency follows from the Pandrosion tangency condition (Theorem 4.1): the gradient of $S(z) = P'(z)$ at m_k is parallel to the side $\alpha_i\alpha_j$, which means the level curve $|z - f_1| + |z - f_2| = 2a$ through m_k has its tangent perpendicular to $\nabla|S|$, hence parallel to the normal of the side—confirming tangency. $\square$

6 Pandrosion influence weights at the foci

Definition 6.1. At a zero f of $S(z) = \sum_k Q(\alpha_k, z)$, the **Pandrosion influence weight** of vertex α_k is:

$$\lambda_k(f) = \frac{|f - \alpha_k|^{-2}}{\sum_{j=1}^3 |f - \alpha_j|^{-2}} \in (0, 1), \quad \sum_k \lambda_k = 1.$$

Proposition 6.2 (Barycentre at foci). *Each focus is the barycentre of the vertices with Pandrosion weights:*

$$f = \lambda_1(f) \alpha_1 + \lambda_2(f) \alpha_2 + \lambda_3(f) \alpha_3.$$

The focus closer to a vertex has a higher weight for that vertex.

Proof. This is the Gauss–Lucas convex combination (Paper 14, Theorem 2.1) specialised to $d = 3$. $\square$

Remark 6.3 (Physical interpretation). The foci are *equilibrium points* of the Pandrosion influence field. Each vertex α_k exerts a “gravitational” pull of strength $|z - \alpha_k|^{-2}$ on a test point z . The foci are the two points where these three pulls balance: $\sum_k Q(\alpha_k, f)/(f - \alpha_k) = 0$, or equivalently $\sum_k (f - \alpha_k)^{-1} = 0$. The Steiner inellipse is the “zero-net-force” locus in this gradient field.

7 The Steiner inellipse parameters

Theorem 7.1 (Pandrosion formulas for the Steiner inellipse). *Let a, b be the semi-axes and e the eccentricity. Then:*

$$c = \frac{|\sqrt{D}|}{3} = \frac{1}{3} \left| \frac{(\alpha_1 - \alpha_2)^2 + (\alpha_1 - \alpha_3)^2 + (\alpha_2 - \alpha_3)^2}{2} \right|^{1/2}, \quad (7)$$

$$2a = |v_k - \sqrt{D}/3| + |v_k + \sqrt{D}/3| \quad (\text{for any } k), \quad (8)$$

$$b = \sqrt{a^2 - c^2}, \quad e = c/a. \quad (9)$$

The axis direction of the ellipse is $\arg(\sqrt{D})$.

Remark 7.2. For an equilateral triangle ($|\alpha_1 - \alpha_2| = |\alpha_1 - \alpha_3| = |\alpha_2 - \alpha_3| = \ell$), $D = 3\ell^2/2$ (if real), $c = \ell\sqrt{6}/6$, and the Steiner inellipse is the incircle. The two foci coincide with the centroid: $f_1 = f_2 = G$, giving $c = 0$ —a circle of radius $b = a$.

8 Extension to degree four

Proposition 8.1 (Quartic critical points via Pandrosion). *For $P(z) = \prod_{k=1}^4 (z - \alpha_k)$, the three critical points (zeros of $S(z) = \sum_k Q(\alpha_k, z) = 4z^3 - 3\sigma_1 z^2 + 2\sigma_2 z - \sigma_3$) are each a convex combination of the four vertices with influence weights $\lambda_k = |f - \alpha_k|^{-2} / \sum |\dots|^{-2}$.*

Remark 8.2. For degree four, there is no single “inscribed ellipse” through the critical points. However, the three critical points form a triangle whose vertices are Pandrosion barycentres of the four roots. This is a “secondary Marden structure”: the critical triangle is contained in the convex hull (by Gauss–Lucas) and its geometry is determined by the Pandrosion influence equilibrium.

9 Conclusion

The Pandrosion quotient provides a transparent, derivative-free path to Marden’s theorem. The key identities are:

Identity	Meaning
$S(z) = \sum Q(\alpha_k, z) = 0$	Foci = influence equilibria
$D = \frac{1}{2} \sum (\alpha_i - \alpha_j)^2$	Discriminant = side geometry
$Q(\alpha_k, m_k) = -\frac{(\alpha_i - \alpha_j)^2}{4}$	Tangency condition
$f = \sum \lambda_k \alpha_k$	Foci as barycentres

The tangency identity $Q(\alpha_k, m_k) = -(\alpha_i - \alpha_j)^2/4$ is, to our knowledge, new. It expresses a deep geometric fact—the Steiner tangent point—as a simple evaluation of a root product.

References

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